

INDIAN ECONOMY

LEVEL
II



LECTURES

ENERGY, MINES & MINERALS ...
AN OVERVIEW OF THE RECENT
DEVELOPMENTS - PART - I

MODULE - 43



INDIA'S ENERGY SECTOR IS GOVERNED BY SEVERAL MINISTRIES/REGULATORS!



- Five different ministries along with several regulators govern India's energy sector.
- There are separate ministries / departments for Petroleum and natural gas, coal, renewable energy and nuclear energy.
- We also have a Ministry of Power, along with State-level bodies that regulate electricity distribution companies, or DISCOMS.
- The petroleum and natural gas sector has two regulators - Directorate General of Hydrocarbons for upstream activities and the Petroleum and Natural Gas Regulatory Board for downstream activities.

IMPORT DEPENDENCE OF CRUDE OIL WAS ENVISAGED TO BE 67% BY 2022 ... BUT IT WENT UP TO 85%!

- In 2015, the Prime Minister, when inaugurating the **Urja Sangam 2015** had urged all stakeholders to increase the domestic production of crude oil and natural gas to reduce import dependence of crude oil from 77% to 67% by 2022, when India celebrates its 75 years of Independence.
- He also said that the import dependence should be brought down to 50% by 2030.



WHAT IS THE REALITY NOW?

As per the Petroleum Planning and Analysis Cell (PPAC), look at India's oil import dependence!

- 2019-20 ... 85 per cent
- 2020-21 ... 84.4 per cent
- 2021-22 ... 85.6 per cent
- 2022-23 ... expected to be around 86 per cent



WHAT ARE THE PROBABLE SOLUTIONS TO REALIZE THE GOAL OF THE PRIME MINISTER?

1 MORE INVESTMENTS IN EXPLORATION



- As per experts, the only way we can reduce our dependence on imports is to increase the size of India-owned exploration and production assets overseas, that is what China has done.
- Our demand for oil is expected to increase substantially as we go for 8% - 10% GDP growth.

From exploration to production — there is long gestation period and oil and gas business requires huge capital!

WHAT ARE THE PROBABLE SOLUTIONS TO REALIZE THE GOAL OF THE PRIME MINISTER?

2

STRENGTHEN THE EXISTING POLICIES



To reduce import dependence, the policymakers have to ensure that all stakeholders are on board and there is no policy flip-flop!

- Public sector oil giant **ONGC** is taking various steps to increase the production by **redevelopment** of existing **matured fields** and development of new/marginal fields.
- Improved oil recovery and **enhanced oil recovery technologies** are being inducted for enhancing recovery from matured fields.
- The Government has introduced various policies for increasing production of oil and natural gas under the **Production Sharing Contract Regime**, **Discovered Small Fields Policy**, **Hydrocarbon Exploration and Licensing Policy (HELP)**, etc.

THREE COMPONENTS OF POWER SECTOR

GENERATION



TRANSMISSION



DISTRIBUTION



GLANCE AT KEY 'UDAY' TARGETS SET IN 2015



AGGREGATE TECHNICAL & COMMERCIAL LOSS

It is calculated as the difference between the energy input in the distribution network and revenue collected for the same.

- Reduction in Aggregate Technical and Commercial losses to 15% by March, 2019.
- Reduction in the gap between per unit Average Cost of Supply and Average Revenue Realised to nil by March, 2019.
- Tariff revisions by DISCOMs.
- Several operational efficiency indicators to be monitored, such as smart metering and feeder segregation etc.

Overall, UDAY was proved to be a failure!

PRESENT COAL CRISIS ... AN EXAMPLE OF UNINTENDED CONSEQUENCES DUE TO THE POLICY DECISIONS!

SEQUENTIAL DEVELOPMENTS SINCE 2000

1 THOUGHT PROCESS AFTER THE REFORMS



- After the reforms of 1991, getting private investment into sectors, where only the public sector existed, became a goal.
- It was thought that with private investments, capacity constraints would be overcome.
- The private sector was also ipso facto supposed to be more efficient.
- The NDA Government at the turn of this century introduced a piece of legislation in Parliament for private coal mining, then developed cold feet and did not pursue it.

PRESENT COAL CRISIS ... AN EXAMPLE OF UNINTENDED CONSEQUENCES DUE TO THE POLICY DECISIONS!

SEQUENTIAL DEVELOPMENTS SINCE 2000

2 LIBERAL ALLOCATION OF COAL BLOCKS

- Later, the UPA government which followed in 2004, took the pragmatic view that private coal mining through legislative change to ensure auctions was not required. The same net result could be achieved by liberally allocating coal blocks for captive use.
- Thermal power plants and other industries - such as cement and steel, were given coal blocks for the asking.
- These were coal blocks which had been investigated, reserves established and were in the pipeline for mining by Coal India.

- As these were given for captive use, the pipeline of new mines for Coal India was reduced.



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SEQUENTIAL DEVELOPMENTS SINCE 2000

2 LIBERAL ALLOCATION OF COAL BLOCKS

GOVERNMENT VIEWPOINT WHILE MAKING LIBERAL ALLOCATIONS

Captive mining would lower costs and liberal allocation in a competitive industry structure would result in “benefits of lower costs” being passed on to downstream users and increasing their competitiveness.



COUNTER VIEW FAULTED THE GOVERNMENT DECISION!

A different view within the government then was that price maximization through auctions should be the way to allocate scarce resources.



PRESENT COAL CRISIS ... AN EXAMPLE OF UNINTENDED CONSEQUENCES DUE TO THE POLICY DECISIONS!

SEQUENTIAL DEVELOPMENTS SINCE 2000

3

CAG AND THE SUPREME COURT



These allocated coal blocks with reserves of about 28 billion tonnes could have achieved a production capacity of at least 300-500 million tonnes per annum by now and coal supply would not have been a constraint today!

- The price maximization view was picked up by the then CAG in a public campaign mode with the use of the term “presumptive loss.”
- The presumed loss figure of Rs.10.67 lakh crore declared by the CAG naturally created a storm. The matter went to the Supreme Court.
- The Government was unable to successfully defend its policy.
- The Supreme Court cancelled the allocation of 214 (all but four) coal blocks in 2014.

PRESENT COAL CRISIS ... AN EXAMPLE OF UNINTENDED CONSEQUENCES DUE TO THE POLICY DECISIONS!

SEQUENTIAL DEVELOPMENTS SINCE 2000

4 MASSIVE INCREASE IN THE THERMAL POWER PLANTS



With the increase in the international coal prices, the thermal power plants which are designed to run with imported coal became uncompetitive!

- In the meantime, thermal power investments by the private sector grew rapidly and this gave India adequate generating capacity at the national level for the first time.
- Faced with inadequate domestic coal production, imports became unavoidable.
- Some private plants, which came up were designed to use only imported coal. This seemed feasible initially as the price of thermal coal in the international market had been low and stable.

PROBLEMS WITH THE COAL SECTOR POWER GENERATION

5 Challenges from increase in solar/wind power, with future probable storage facilities.

4 Reduction in demand, due to energy conservation and efficiency.

3 Decline in Plant Load Factor further stresses the finances of coal based power plants.

2 Lack of flexibility for integrating with the grid (flexibilization), instantly as and when required, is another problem.

COAL IS HERE TO STAY ... BUT CONCOMITANT CHALLENGES LIE AHEAD!

1 Coal is the dirty fuel. Pollutants is the real problem, mainly by the power plants, which were established decades ago. For the existing power plants, satisfying the pollution norms deadline, is extended beyond 2022.

PLANT LOAD FACTOR

Plant Load Factor is the ratio between the actual energy generated by the plant to the maximum possible energy that can be generated with the plant working at its rated power capacity for a given duration.

LOGISTICS ISSUES IN INTEGRATION TO THE GRID

COAL
BASED



NUCLEAR
POWER



GAS
BASED



STORAGE
HYDRO



*Making these
two flexible in
operations is
the real
problem!*

*These are Flexible
and can easily be
integrated, as and
when required*



SOLAR
POWER

*The power generation
is intermittent, but we
have to take entire
power generated to the
grid, as and when it is
produced.*



WIND
POWER

NATIONAL GAS GRID

- With the aim to create nationwide gas grid, Petroleum and Natural Gas Regulatory Board (PNGRB) has authorized approximately 33,764 km natural gas pipeline network across the country, as on 31.03.2021.
- The National Gas Grid would connect all major demand and supply centres in India.
- This would ensure easy availability of natural gas across all regions and also potentially help to achieve uniform economic and social progress.

- In order to develop the National Gas Grid, the Government has taken a decision to provide a capital grant of Rs.5,176 crore under Pradhan Mantri Urja Ganga in 2016 for development of a 2,655 Km long Jagdishpur-Haldia - Bokaro- Dhamra Gas Pipeline (JHBDPL) project, which passes through 50 districts in the States of Uttar Pradesh, Bihar, Jharkhand, Odisha & West Bengal.



250 MILLION SMART METERS ARE PLANNED

FUNCTIONS OF SMART METERS

- They have in-built features such as bi-directional communication, and net metering.
- There are options for prepaid and post-paid, and time-of-day tariff features.
- With smart meters, a discom can manage its options better.



WHAT ARE THE BENEFITS CITED BY EESL?

- EESL's National Smart Meter Programme Dashboard reveals tremendous benefits:
 - ✓ 11-36% reduction in Aggregate Technical and Commercial (AT&C) losses.
 - ✓ 21% improvement in billing efficiency or Rs. 301 per month per meter (though experts dispute this).



CHALLENGES IN SCALING UP TO 250 MILLION SMART METERS

1 CONSUMER-CENTRIC FEATURES

The **disconnection** of power connection should not be **automatic** for the **prepaid meters**. There should be **sufficient time**, which must be given to the **consumers**, as what is being done in the present state of **traditional meters**.



2 PRIVACY

- High resolution **smart meter data**, through **analysis** and **inference**, has the potential to reveal personal information of consumers such as **occupancy**, appliance usage patterns and even sensitive information such as **entertainment preference**.
- Hence, **data privacy issues** must be taken care of.

3 RIGHTS TO THE CONSUMERS

- When the consumer is going for smart meters, they should see some **tangible benefits** from smart meters.
- We should think about **automatic compensation** for power outages exceeding certain period of time.

Assurance of steady supply and compensation for outages is one good way to get consumers' buy-in for smart metering!

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